

Abhishek H R

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SUMMARY

Software Developer with a strong foundation in backend engineering, cloud deployment, and applied AI, currently researching AI account infrastructure — a universal layer for safe, autonomous, cross-platform AI action. Skilled in Java Spring Boot, REST APIs, and LLM integration. Passionate about building AI systems that are accountable, auditable, and enterprise-ready, with a focus on AI safety principles including rollback, attribution, and governance.

CURRENT RESEARCH / PROJECT

AI Account Infrastructure

2026 – Present

Independent Research — AI Systems, Safety & Autonomous Agents

- Designing a universal AI account layer enabling cross-platform autonomous actions (email, tasks, calendar) across Claude, ChatGPT, and Gemini without per-workflow configuration.
- Researching AI-to-AI communication protocol where two users' AI agents negotiate, coordinate, and act on each other's behalf — a new human-AI collaboration paradigm.
- Building unified memory architecture syncing context across AI platforms, with rollback and audit systems to log and reverse autonomous actions — addressing a critical AI safety gap.
- Designing enterprise governance layer with permission controls, action attribution, and compliance-ready audit trails for safe, accountable AI deployment at scale.

EXPERIENCE

Technical Intern — Backend Engineering

Sep 2024 – Dec 2024

Ugyan Edutech Pvt Ltd, Bengaluru, Karnataka

- Designed and tested RESTful APIs in Java Spring Boot for scalable backend services.
- Integrated relational databases with CRUD pipelines; optimised data storage and retrieval performance.
- Debugged and deployed Spring Boot applications to production, improving system reliability and uptime.

PROJECTS

Desktop AI Assistant

Python, OpenAI API

- Voice-controlled assistant using OpenAI API to dynamically generate and execute task-specific code with safe auto-deletion — early agentic AI pattern.

E-Commerce Backend System

Java, Spring Boot, JWT, PostgreSQL

- Modular backend with JWT auth, paginated APIs, and integrated order/payment workflows.

FlowNotes — Chrome Extension

JavaScript, Chrome APIs

- Production Chrome extension for site-specific notes; published to Chrome Web Store with privacy-first local storage.

Smart Sprout — IoT Monitoring

ESP32, Firebase, React Native

- Real-time soil monitoring with solar power, Firebase backend, and React Native mobile app.

TECHNICAL SKILLS

Languages: Python, Java, C, C++

AI & LLM: OpenAI API, Claude API, LLM integration, Agentic systems, Prompt engineering

AI Safety: Audit trails, action attribution, rollback systems, human-in-the-loop, governance frameworks

Backend & Cloud: Spring Boot, REST APIs, JWT, Docker, AWS, RDS

Databases: PostgreSQL, MySQL, MongoDB

Tools & Libraries: Git, GitHub, Postman, NumPy, Pandas, Scikit-learn, React Native

EDUCATION

B.E. Computer Science & Engineering — CGPA: 9.20

Sep 2021 – Jun 2025

Visvesvaraya Technological University | Navkis College of Engineering, Hassan, Karnataka

CERTIFICATIONS

- IoT Systems Design
- Data Structures and Algorithms
- Full Stack Java Development